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1. Overview

In accordance with 21 CFR section 807.81, it was determined that a premarket notification submission is required for the ARTSENS Plus PWV device. FDA final guidance “510(k) Program: Evaluating Substantial Equivalence in Premarket Notifications [510(k)], dt: JULY 2014” was referred to in creating this document.

ARTSENS Plus PWV device is substantially equivalent to the following predicate devices:

	Predicate Device	Manufacturer	510(k)No.	Date Cleared
Primary Predicate	SphygmoCor Xcel	AtCor Medical Pty Ltd	K122129	November-16-2012
Secondary Predicate	Vscan Air	GE Medical Systems Ultrasound and Primary Care Diagnostics, LLC	K202035	October-19-2020

2. Regulatory Information

The comparison of the indications for use, technological characteristics – functional, hardware, software and general, and safety and effectiveness of subject device and predicate devices, demonstrated that ARTSENS Plus PWV was similar to SphygmoCor XCEL and Vscan Air, and hence the substantial equivalence is established as follows;

Product Name	Subject Device	Primary Predicate Device	Secondary Predicate Device
Trade/Brand Name	ARTSENS Plus PWV	SphygmoCor XCEL	Vscan Air
Risk classification	II	II	II
US FDA Product code	ITX (DXN, DSK)	DXN(DSK)	ITX (IYN, IYO)
US FDA Regulation	892.1570	870.1130	892.1570

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3. Intended Use Comparison

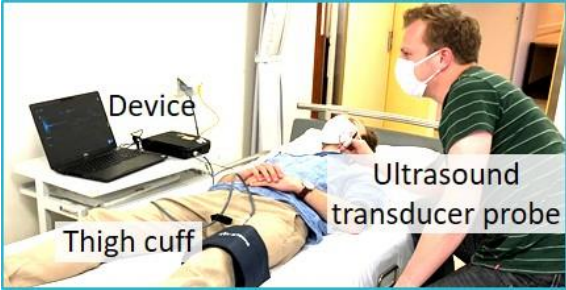
3.1 Comparison with Primary predicate

Subject Device	Primary Predicate Device	Establishment of Equivalence
The ARTSENS Plus PWV, Pulse Wave Velocity (PWV) device is intended to obtain PWV measurements. The device enables visualization and measurement of anatomical structures and fluid, including blood flow. It is to be used on those patients where information related to aortic PWV is desired but the risks of cardiac catheterization procedure or other invasive monitoring may outweigh the benefits. The device is used on adults 18 years of age and older patients only.	The SphygmoCor XCEL System provides a derived ascending aortic blood pressure waveform and a range of central arterial indices. These measurements are provided non-invasively through the use of a tonometer and a brachial cuff. It is to be used on those patients where information related to ascending aortic blood pressure is desired but the risks of cardiac catheterization procedure or other invasive monitoring may outweigh the benefits. Additionally, the SphygmoCor XCEL System automatically measures Systolic blood pressure and Diastolic blood pressure. The SphygmoCor XCEL Pulse Wave Velocity (PWV) option is intended to obtain PWV measurements. The PWV option is used on adult patients only.	The bolded text on the subject device and primary predicate device show the Substantial Equivalence correlation. We do not claim compliance to the rest of the un-bolded text. The equivalence is demonstrated by the following similarities: - Both devices intend to measure aortic PWV. - Both devices record blood pulse signals from same anatomical locations – carotid and femoral, using a neck probe and thigh cuff. - Both are designed to calculate PWV non-invasively, providing lower-risk alternatives to the invasive arterial line catheterization PWV approach. - Both are intended to be used on adults only. The primary predicate SphygmoCor XCEL System has two modes of operation - Pulse wave analysis mode (PWA), where it provides a derived ascending aortic blood pressure waveform, a range of central arterial indices and blood pressure from arm. - PWV mode for aortic PWV measurement.
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		The subject device claims Substantial Equivalence with SphygmoCor XCEL to its PWV measurement mode alone, where both devices use a probe on neck and cuff on thigh to provide non-invasive PWV. Further, any measurement done by predicate device in PWA mode does not affect the measurements in the PWV modes
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3.2 Subject and Primary Predicate device illustrations

Subject Device	Primary Predicate Device	Establishment of Equivalence
		Subject device's operation methodology is identical to the SphygmoCor XCEL System's in PWV mode, where operator uses a probe on neck and a cuff on thigh to measure the aortic PWV.

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		Subject device's components are identical to the SphygmoCor XCEL System's in PWV mode.
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3.3 Comparison with Secondary predicate

Subject Device	Secondary Predicate Device	Establishment of Equivalence
The ARTSENS Plus PWV, Pulse Wave Velocity (PWV) device is intended to obtain PWV Pulse Wave Velocity (PWV) measurements through a combination of pulse waves acquired from an ultrasound transducer probe and inflatable cuff. The device enables visualization and measurement of anatomical structures and fluid, including blood flow. It is to be used on those patients where information related to aortic PWV is desired but the risks of cardiac catheterization procedure or other	Vscan Air is a battery-operated, software-based, general-purpose ultrasound imaging system for use by qualified and trained healthcare professionals or practitioners legally authorized or licensed by law in the country, state, or local municipality where they practice. Users may or may not work under the supervision or authority of a physician. Users may also include medical students working under the supervision or authority of a physician during their education/training. The device enables visualization and measurement of anatomical structures and fluid, including blood flow. Vscan Air's pocket-sized portability and simplified user interface enable integration into training sessions and examinations in professional healthcare facilities (e.g., hospital, clinic, medical office),	The bolded text on the subject device and secondary predicate device show the Substantial Equivalence correlation. We do not claim compliance to the rest of the un-bolded text. The subject device with its ultrasound transducer is enabling visualization and measurement of carotid pulse waves, as a result of dynamics in anatomical structures and fluid including blood flow. Both devices use ultrasound as their primary mode of action for capturing data and displaying that data.

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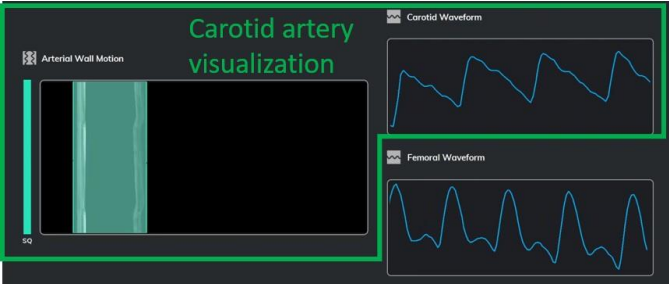
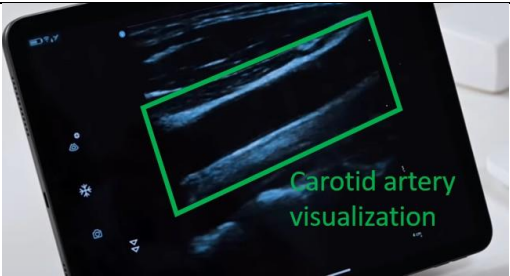
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invasive monitoring may outweigh the benefits. The device is used on adults 18 years of age and older patients only.	home environments, road/air ambulances, and other environments described in the user manual. The information can be used for basic/focused assessments and adjunctively with other medical data for clinical diagnosis during routine, periodic follow-ups, and triage. Vscan Air supports Black/White (B-mode), Color Flow (Color Doppler), Combined (B + Color Doppler), and Harmonic imaging modes with both curved and linear array transducers. With the curved array transducer of the dual-headed probe solution, specific clinical applications and exam types include abdominal, fetal/obstetrics, gynecological, urology, thoracic/lung, cardiac (adult and pediatric, 40 kg and above), vascular/peripheral vascular, musculoskeletal (conventional), pediatrics, and interventional guidance (including freehand needle/catheter placement, fluid drainage, nerve block, and biopsy). With the linear array transducer of the dual-headed probe solution, specific clinical applications and exam types include vascular/peripheral vascular, musculoskeletal (conventional and superficial), small organs, thoracic/lung, ophthalmic, pediatrics, neonatal cephalic, and interventional guidance (including freehand needle/catheter placement, fluid drainage, nerve block, vascular access, and biopsy).	The primary difference in the visualization is that the subject device measures carotid pulse waves as the dynamics in the artery's distension and displays the data as artery motion-mode animation and carotid artery pulse waves. The secondary predicate device V-scan air illustrates a B-mode image.
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3.4 Subject and Secondary Predicate device illustrations

Subject Device	Secondary Predicate Device	Establishment of Equivalence
	 	Both the devices use ultrasound probes with ultrasonic piezo-crystal elements. Similar to subject device, the predicate is also used for carotid application. The subject device uses one element, and the predicate devices uses an array of elements.
		The primary difference in the visualization is that the subject device measures carotid pulse waves as the dynamics in the artery's distension and displays the data as artery motion-mode animation and carotid artery pulse waves. The secondary predicate device V-scan air illustrates a B-mode image.

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4. Technological Characteristics Comparison with Primary Predicate

4.1 Comparison with Secondary predicate			
	Subject Device (ARTSENS Plus PWV)	Primary Predicate (SphygmoCor XCEL)	Equivalence comparison
Rx or OTC	Prescription Use (Rx)	Prescription Use (Rx)	Equivalent to Primary Predicate
Environment to fuse	Clinical or Research setting	Clinical or Research setting	Equivalent to Primary Predicate
Target population	Adult	Adult	Equivalent to Primary Predicate
Product Type	Hardware + Software 1) Table top Hardware unit that is to be connected to a host computer via USB cable. 2) Software for signal acquisition & processing, visualization, and storage.	Hardware + Software 1) Table top Hardware unit that is to be connected to a host computer via USB cable. 2) Software for signal acquisition & processing, visualization, and storage.	Equivalent to Primary Predicate
4.2 Functional features			
Measurements supported	Aortic PWV	SphygmoCor XCEL PWV mode is intended to obtain aortic PWV.	Equivalence is claimed for the primary predicate's PWV mode; both devices commonly measure the Aortic PWV by applying a similar technique and calculations. The primary predicate device also measures additional parameters (ascending aortic blood pressure waveform and a range of central arterial indices) in PWA mode to which equivalence is not claimed. It may be noted that the Aortic PWV calculation is not dependent on any of

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			these additional parameters that are measured only by the predicate device and not measured by the subject device. Therefore, in terms of the safety and efficacy of the PWV measurement the subject device equivalent to the primary predicate. Subject device was compared to the predicate for performance on human subjects, demonstrating a strong agreement.
PWV range	2–25m/s	2–25m/s	Equivalent to Primary Predicate
Signals captured for PWV	Blood pulse waves are captured from carotid and femoral arteries by the electronic device	Blood pulse waves are captured from carotid and femoral arteries by the electronic device	Equivalent to Primary Predicate
Signal acquisition manner	Carotid and femoral blood pulse waves are simultaneously recorded, and the acquisition is time-synchronized	Carotid and femoral blood pulse waves are simultaneously recorded, and the acquisition is time-synchronized	Equivalent to Primary Predicate
Method for PWV distance measurement	Calculated as 80% of the direct distance from the carotid artery to the top edge of the femoral cuff	Calculated as 80% of the direct distance from the carotid artery to the top edge of the femoral cuff when used in direct method.	Equivalent to Primary Predicate
Method for Pulse Transit time (PTT)	Delay between the foot of carotid and femoral pulse waves, identified using the Intersecting-tangent method	Delay between the foot of carotid and femoral pulse waves, identified using the Intersecting-tangent method	Equivalent to Primary Predicate
Calculation of Aortic PWV	Calculated as the PWV distance divided by the pulse transit time.	Calculated as the PWV distance divided by the pulse transit time.	Equivalent to Primary Predicate

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4.3 Hardware features

Key components for PWV measurement	Acquisition unit + carotid pulse acquisition probe + Thigh cuff	Acquisition unit + carotid pulse acquisition probe + Thigh cuff	Equivalent to Primary Predicate
Interface	The device is interfaced to a host computer via USB, where the software is installed for processing, display, and storage	The device is interfaced to a host computer via USB, where the software is installed for processing, display, and storage.	Equivalent to Primary Predicate
Carotid Pulse wave acquisition	Device uses an ultrasound transducer probe, utilizing a single element piezo-crystal, for capturing carotid pulse waves as artery's anatomical distension.	Device uses a tonometer transducer that provides an uncalibrated carotid pressure pulse wave.	Equivalence with predicate is claimed with respect to usage of carotid probe for carotid blood pulses. Note that PWV measurement doesn't rely on the amplitude of the pulse wave but just the morphology of the pulse wave. Now, as the pressure pulse causes artery to distend, the carotid pulse waves captured as pressure pulse by the predicate is identical to distension pulse by the subject device. It may be further noted that the use of ultrasound transducer probe by the subject device for capturing carotid artery's anatomical distension is equivalent to the secondary predicate V-scan Air. More characteristics pertinent to ultrasound transducer probe are compared in section 5.
Femoral pulse wave acquisition	An uncalibrated femoral pressure pulse wave is captured using an inflated bladder-type cuff wrapped around the thigh and a pressure sensor	An uncalibrated femoral pressure pulse wave was captured using an inflated bladder-type cuff wrapped around the thigh and a pressure sensor	Equivalent to Primary Predicate

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Cuff location	Upper thigh	Upper thigh for “PWV mode” measurement. Arm for “PWA mode” measurement.	Equivalence is claimed for only the “PWV mode” of the predicate device. The subject device doesn’t do blood pressure measurement at all, which the predicate device does in “PWA mode”. Hence subject device doesn’t come with an Arm cuff
Applied parts Material	Carotid probe: ABS Plastic & Silicon Thigh cuff: Cuff covers are made of Nylon and Velcro.	Carotid probe: Tecason & Silicon Thigh cuff: Cuff covers are made of Nylon and Velcro.	The applied parts serve an equivalent intended purpose of capturing simultaneous carotid and femoral pulse waves. The carotid probe has a different material than used for the predicate device, ABS plastic versus Tecason. However, ABS plastics are recognized by the FDA as one of the materials that pose a very low biocompatibility risk because they have a long history of safe use in legally marketed medical devices. <u>In addition, for all the applied parts the biocompatibility safety tests performed as per ISO 10993-1, confirmed that the applied parts of the subject device are equivalently safe to use.</u>
Calibration	The OEM pressure module uses a pressure transducer to monitor and maintain the inflation pressure at a set point for femoral pulse wave recording. Annual calibration is hence required for the OEM pressure module to	The OEM pressure module uses a pressure transducer to monitor and maintain the inflation pressure at a set point for femoral pulse wave recording. Annual calibration is hence required for the OEM pressure module to	Equivalent to Primary Predicate

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	ensure reliability in this regard.	ensure reliability in this regard.	
Electrical power	Can be powered by AC mains and also battery operated	Can only be powered by mains	Equivalent to Primary Predicate in terms of its mains power operation. The subject device uses a battery backup as well, which is advantageous during power outages and lack of immediate access to main power. The battery has been tested for safety as per standard IEC 62133-2, and the subject device is safe and effective for use, equivalent to the predicate device.
4.4 Software features			
Software graphic user interface	Assists operator to initiate measurements, displays measured waveforms and resulting parameters	Assists operator to initiate measurements, displays measured waveforms and resulting parameters	Equivalent to Primary Predicate
Patient information	Offers provision for patient details entry	Offers provision for patient details entry	Equivalent to Primary Predicate
Acquisition and processing for PWV measurements	Acquires digitized signals from the device to the host computer and processes for final measurements	Acquires digitized signals from the device to the host computer and processes for final measurements	Equivalent to Primary Predicate

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Visual feedback	Provides operator the visual feedback of 1. Recorded carotid and femoral pulse waves, 2. Signal quality metrics for carotid pulse waves	Provides operator the visual feedback of 1. Recorded carotid and femoral pulse waves, 2. Signal quality metrics for carotid pulse waves	Equivalent to Primary Predicate
Cuff inflation	Controls the thigh cuff inflation and deflation for femoral pulse waves	Controls the thigh cuff inflation and deflation for femoral pulse waves in the PWV mode option of the software	Equivalent to Primary Predicate
Cuff control inspections	Inspects for cuff leaks and over-pressure inflation	Inspects for cuff leaks and over-pressure inflation	Equivalent to Primary Predicate
Data Export	Available, measurement data can be exported as a PDF report	Available, measurement data can be exported as a PDF report	Equivalent to Primary Predicate

4.5 Product general features

Device form factor	Dimension: 240 mm x 170 mm x 60 mm Weight: Approx. 1 Kg	Dimension: 99 mm x 190 mm x 172 mm Weight: Approx. 1 Kg	The subject device is equivalently portable as Primary predicate.
Data transmission	The device is connected to the host computer via a USB cable.	The device is connected to the host computer via a USB cable.	Equivalent to Primary Predicate
Minimum computer requirement	Computer with Windows 10 OS or above, 1xUSB (type A) 2.0 port or higher	Computer with Windows 7, 8 Pro or 10 OS, 1xUSB (type A) 2.0 port or higher	Equivalent to Primary Predicate

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Ports	Connection port for external power supply, interface for connecting carotid probe, and cuff's pneumatic interface to the device.	Connection port for external power supply, interface for connecting carotid probe, and cuff's pneumatic interface to the device.	Equivalent to Primary Predicate
Memory	Data of the measurements are not stored in the device.	Data of the measurements are not stored in the device.	Equivalent to Primary Predicate
Generate a report for user	Yes	Yes	Equivalent to Primary Predicate
Review measurement records	Available	Available	Equivalent to Primary Predicate

5. Comparison of Labels with Primary Predicate

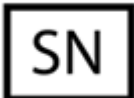
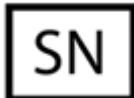
	Subject Device (ARTSENS Plus PWV)	Primary Predicate (SphygmoCor XCEL)	Equivalence comparison
Labels on the device	- A device serial number label - Labels for status indicators—such as power-on, measurement progress, battery charging. - Labels for connection interfaces—cuff, probe, and USB ports	- A device serial number label - Labels for status indicators—such as power-on, measurement progress, cuff connected, and probe connected. - Labels for connection interfaces—cuff, probe, and USB ports	The subject device has equivalent labels as the predicate device. The subject checks and indicates the cuff and probe connection status when the software is started. Therefore, these indications are shown as part of software, and not shown as LEDs on the device. Hence labels for cuff connected, and probe connected are not applicable on the subject device, but in the subject device's software.

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			<u>Likewise, the predicate device doesn't come with a battery backup, therefore the battery status label which is part of subject device is not applicable to predicate device.</u>
Refer to instruction manual/booklet			Both the devices equivalently have a symbol for refer to IFU, the subject device uses a symbol according to ISO graphical symbols (reference number: M002)
Type BF Applied Part			Equivalent to Primary Predicate
Date of Manufacturer			Equivalent to Primary Predicate
Serial Number			Equivalent to Primary Predicate
Manufacturer			Equivalent to Primary Predicate
Probe port			Both devices equivalently represent the probe connection interface, the subject device uses a standard symbol for pencil probes from ISO graphical symbols (reference number: 5707)

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Cuff Indicator	CONNECT CUFF		Both devices equivalently represent the cuff connection interface, but the subject uses text to avoid confusion.
USB port			Equivalent to Primary Predicate
Power port			Both devices equivalently indicate the DC power inlet port, the subject device mentions the input voltage level.
Indicating power on			Both devices indicate power on status. The subject device powers on when “connected to the host” for improved battery management, but the predicate device has a power switch to turn on the device. Hence the symbols used differ in the subject device differs for more appropriate indication, that the device is connected to the host.
Status Indicator			Both devices indicate progression of a measurement, the subject device uses an ISO graphical symbol for representing progress (reference number: 3318)

6. Subsequent Technological Characteristics Comparison with Secondary Predicate

	Subject Device (ARTSENS Plus PWV)	Secondary Predicate (Vscan Air)	Equivalence comparison
Transducer	Suitable	Suitable	Equivalent to Secondary Predicate, as both the devices are suitable for

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probe applicability to peripheral vascular			performing scans on superficial vessels.
Anatomical structure analysis	Yes	Yes	Subject device claims its equivalence to secondary predicate with respect to anatomical structure measurement, more specifically for carotid arteries.
Ultrasound recording feedback	Provides visual feedback for reliable recording	Provides visual feedback reliable recording	Subject device equivalently provides means for visualization during performing a measurement to ensure adequate feedback to operators for reliable capturing of data. However, V-scan air illustrates a B-mode image as feedback, and Subject device illustrates artery motion-mode animation and carotid artery pulse waves.
Number of elements	Transducer probe has one piezo crystal element, as the scanning and processing are in A-mode.	For peripheral vascular applications a linear array transducer with 192piezo crystal elements is used. The scanning and processing are in B-mode	Both the device as substantially equivalent, in using piezo crystal elements. For PWV measurements, and carotid pulse waveforms a single scan line suffices, therefore the subject device doesn't need more than one element. The efficacy of the carotid pulse recording and the PWV measurements has been validated in human subjects by comparing it to the that of primary predicate and attached.
Frequency	Broad bandwidth of 3 - 7 MHz with center frequency of 5 MHz	Broad bandwidth of 3 - 12 MHz with center frequency of 7.7 MHz	The bandwidths and center frequencies of the transducers are similar to each other, making it suitable for 40 mm depth scans with respect to peripheral vascular applications.

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Mode of operation	Pulse echo	Pulse echo	Subject device claims equivalence to the pulse echo mode of operation. Predicate supports other modes of operation, such doppler, as well which are not applicable to the subject device.
Applied part	Biocompatible plastic with adequate grip	Biocompatible plastic with adequate grip	Materials of both the devices are biocompatible and equivalently safe.
Energy delivery	Ultrasound transducer delivers acoustic power	Ultrasound transducer delivers acoustic power	Subject device is equivalently safe to use for vascular applications as that of the secondary predicate. Subject device's ultrasound transducer is compliant to the FDA recognized safety standards • IEC 61161 Edition 3.0 2013-01, Ultrasonics - Power measurement -- Radiation force balances and performance requirements • IEC 60601-2-37 Edition 2.1 2015, Medical electrical equipment - Part 2-37: Particular requirements for the basic safety and essential performance of ultrasonic medical diagnostic and monitoring equipment • IEC 62359 Edition 2.1 2017-09, Ultrasonics - Field characterization - Test methods for the determination of thermal and mechanical indices related to medical diagnostic ultrasonic fields
Acoustic testing as per FDA	Track 1	Track 3	As per the FDA guidance document “Marketing Clearance of Diagnostic Ultrasound Systems and Transducers”, track-1 devices should have I_{SPTA} level restricted to 720 mW cm^{-2} for peripheral vascular applications. Unlike the secondary predicate, the subject device is restricted only for carotid site, qualifying it as peripheral vascular application. Therefore, the subject device also limits the acoustic power delivery to a constant

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			value within the safe limits specified by FDA by not providing any form of software or hardware control that is capable of altering the acoustic output. Therefore, the subject device qualifies for track-1 testing. The subject device’s maximum acoustic output was tested as per the recognized standards, which is way below the track-1 limit specified by the FDA for peripheral vascular application.
Battery backup	Available	Available	Equivalent to Secondary Predicate

7. Conclusive summary

FDA 510(k) decision making Flowchart questions	Response	Comments
Decision 1: Is the predicate device legally marketed?	Yes	NA
Decision 2: Do the devices have the same intended use?	Yes	The subject device claims equivalence to the primary predicate device in the PWV mode of operation, for the measurement of PWV on adult population. Further, the subject device claims equivalence to a secondary predicate Vscan Air for the use of an ultrasound probe for carotid artery analysis
Decision 3: Do the devices have the same technological characteristics?	Yes	The subject device claims substantial equivalence for the majority of the technology characteristics with primary predicate, apart from the usage of an ultrasound transducer probe for the carotid pulse waves. However, subsequently the subject device claims equivalence to a secondary predicate Vscan Air for the use of ultrasound probe for carotid artery analysis.

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Decision 4: Do the different technological characteristics of the devices raise different questions of safety and effectiveness?	No	There are no different technological characteristics for the predicate device since it claims equivalence to two predicate devices one for the PWV recognized technology and ultrasound method for taking such readings.
Decision 5a: Are the methods acceptable	Yes	The safety and effectiveness equivalence are acceptable as they follow the same product codes and regulations of the predicate devices which meet the applicable associated FDA recognized standards.
Decision 5b: Do the data demonstrate substantial equivalence?	Yes	The results of the performance tests based on the applicable standards from the predicate devices demonstrates that the subject device is as safe and effective to use.
Based on navigating the decision-making flowchart questions and answers, it is concluded that the subject device is substantially equivalent to the predicate device and subsequently to the secondary predicate.		

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